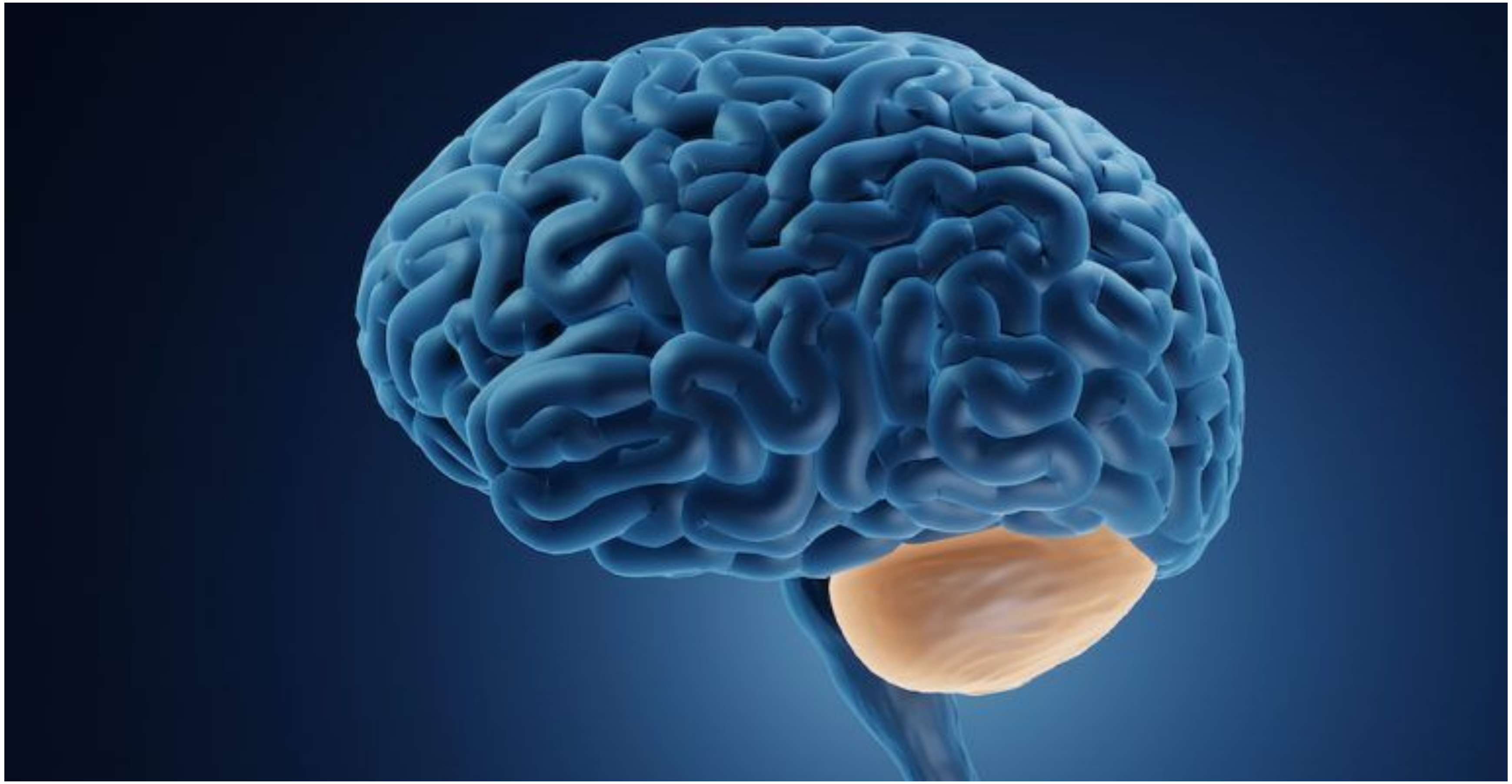
The Cerebellum



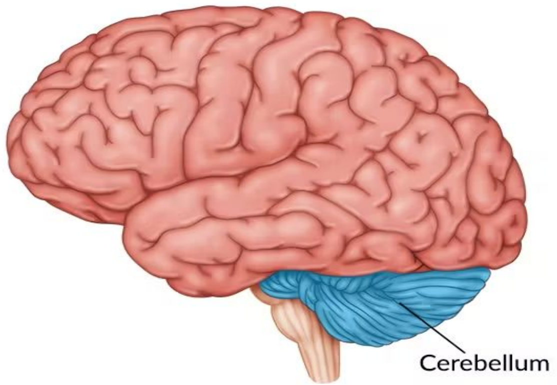
Learning objectives:

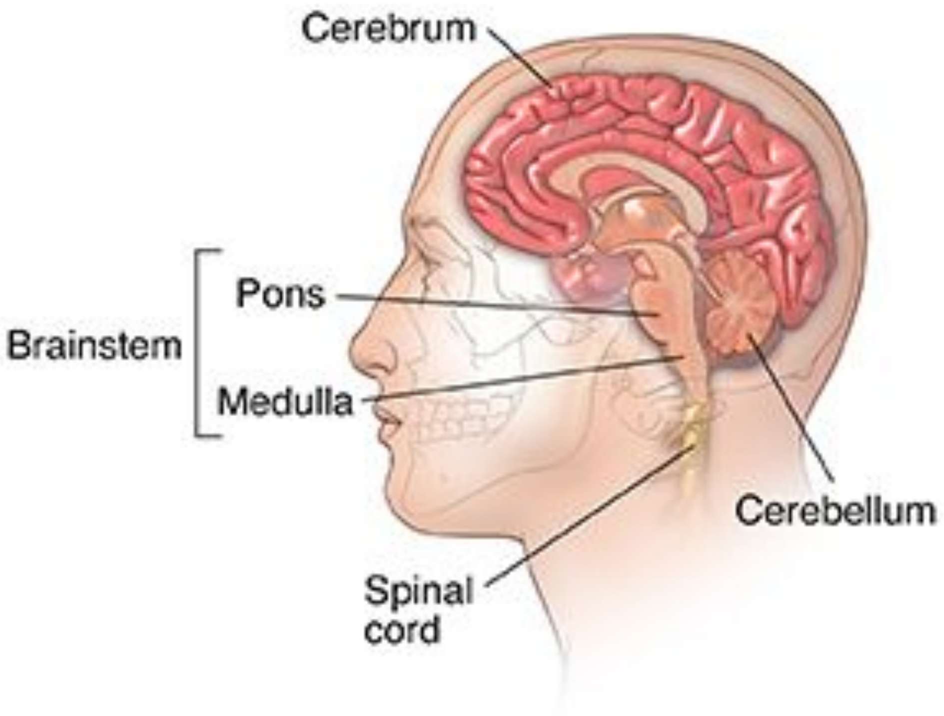
At the end of the lecture, the students should be able to:

- **Describe the gross appearance/external features of cerebellum.**
- **Explain the internal features of cerebellum.**
- **Define the layers of cerebellar cortex and name the prominent cell types present in each layer(Histology).**
- **Define the climbing & mossy fibers.**
- **Enlist the deep cerebellar nuclei.**
- **Name & describe the cerebellar afferent pathways.**
- **Name & describe the cerebellar efferent pathways.**
- **Enlist the functions of cerebellum.**
- **Define some common manifestations of cerebellar dysfunction.**

Introduction

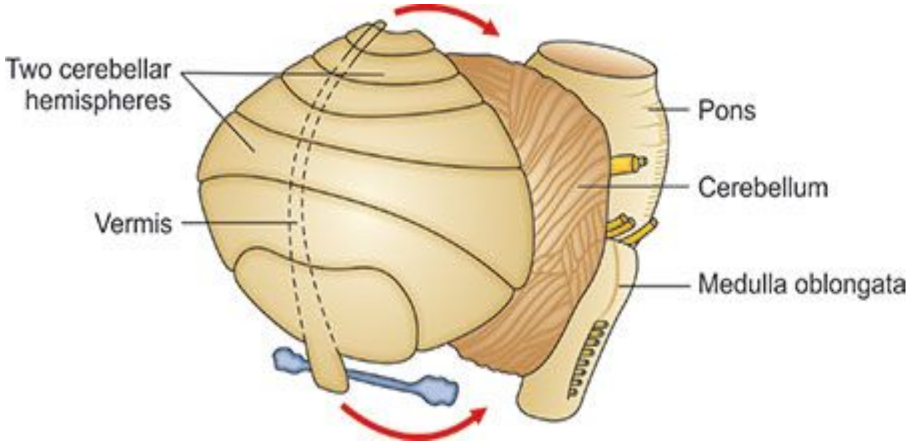
- The cerebellum is the largest part of the hindbrain.
- It plays an important role in the control of posture and voluntary movements.
- It is situated in the posterior cranial fossa.
- Superiorly it is covered by the tentorium cerebelli.
- It lies posterior to the 4th ventricle, pons, and medulla oblongata.

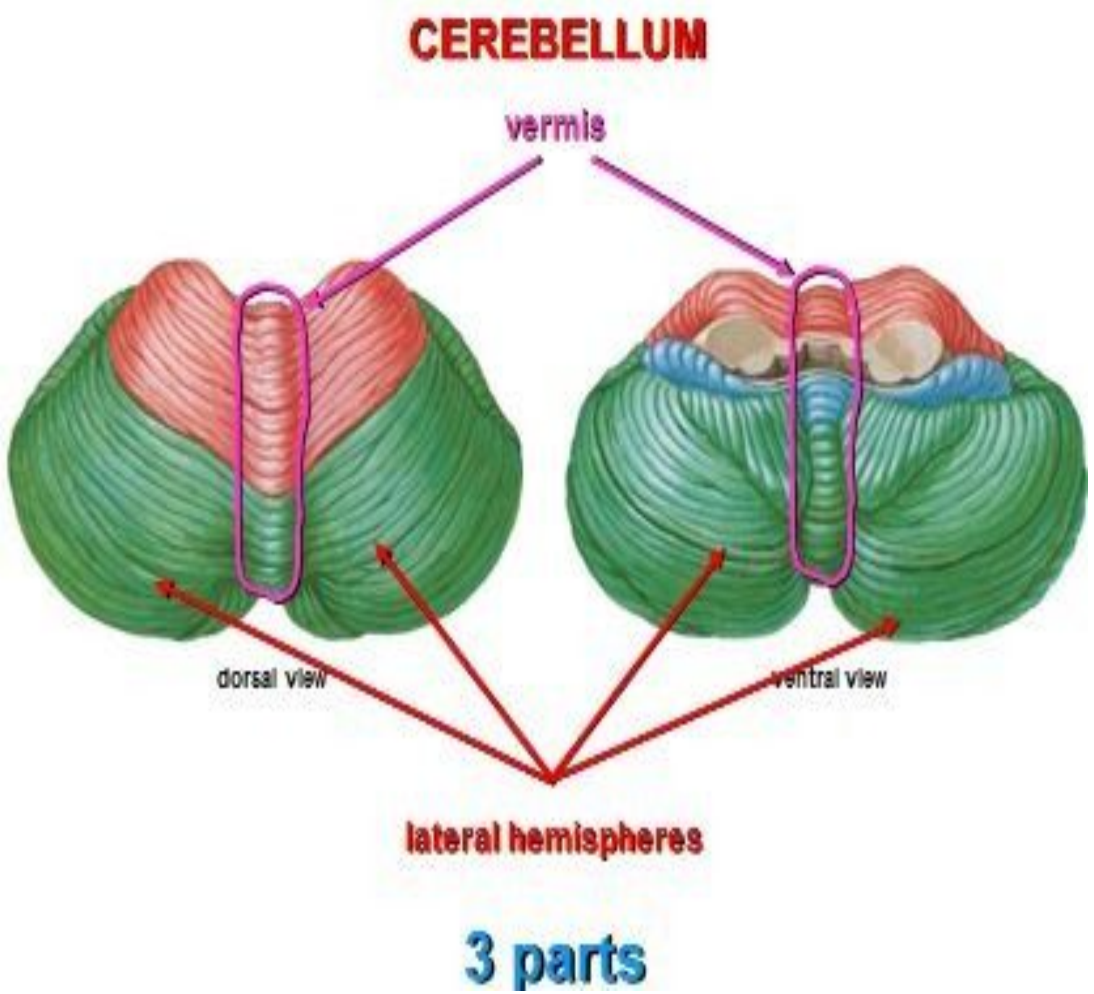




Gross appearance

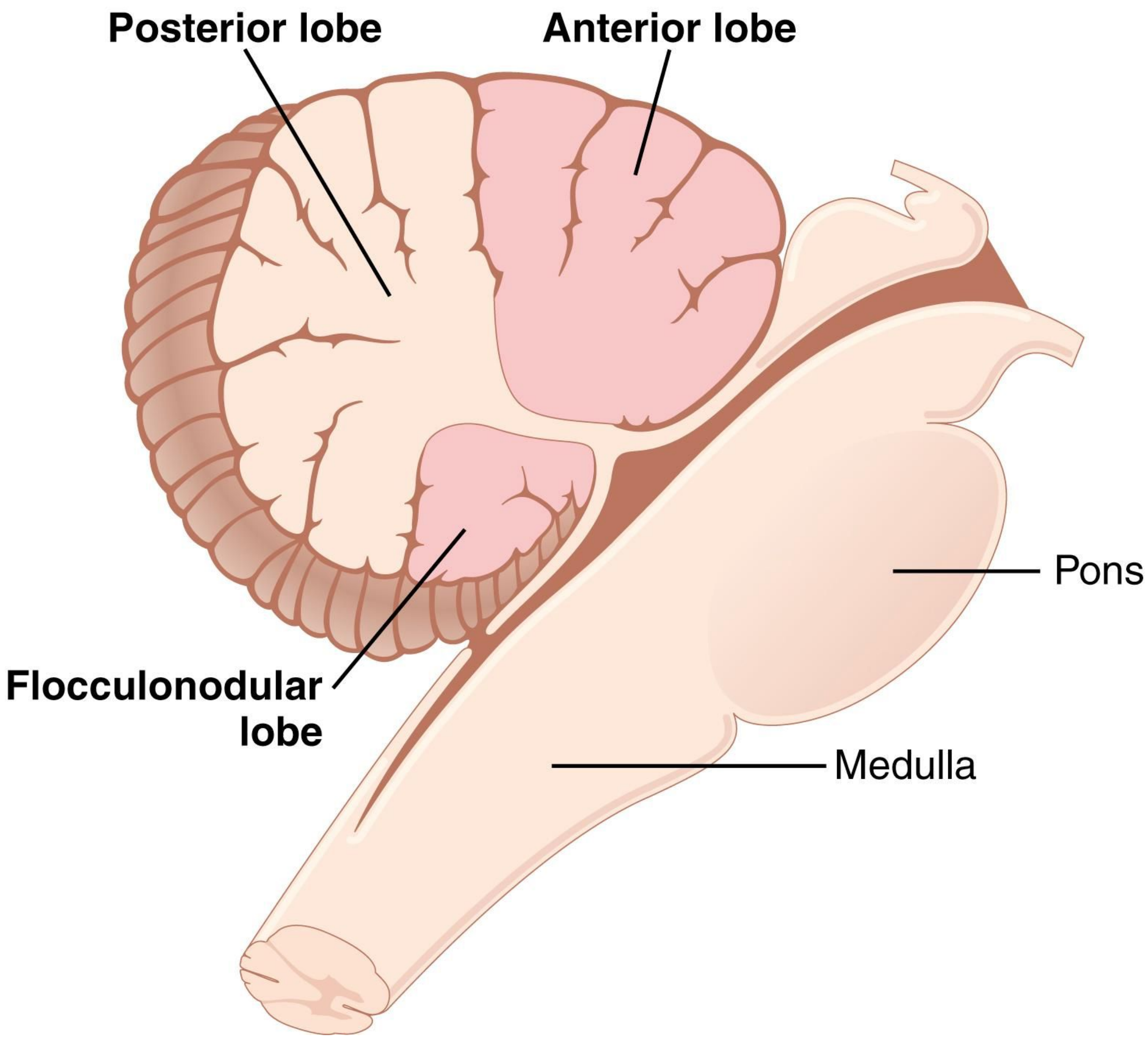
- The cerebellum is somewhat ovoid in shape.
- It consists of the two cerebellar hemispheres.
- The cerebellar hemispheres are joined together by a median constriction called the *vermis*.



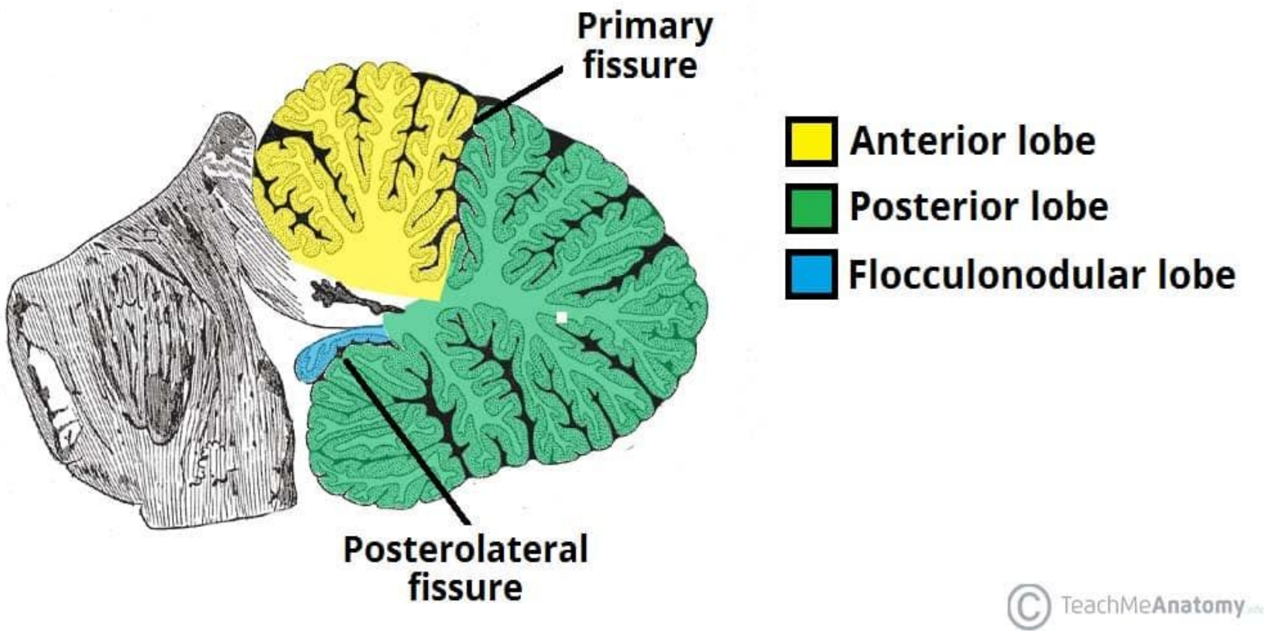


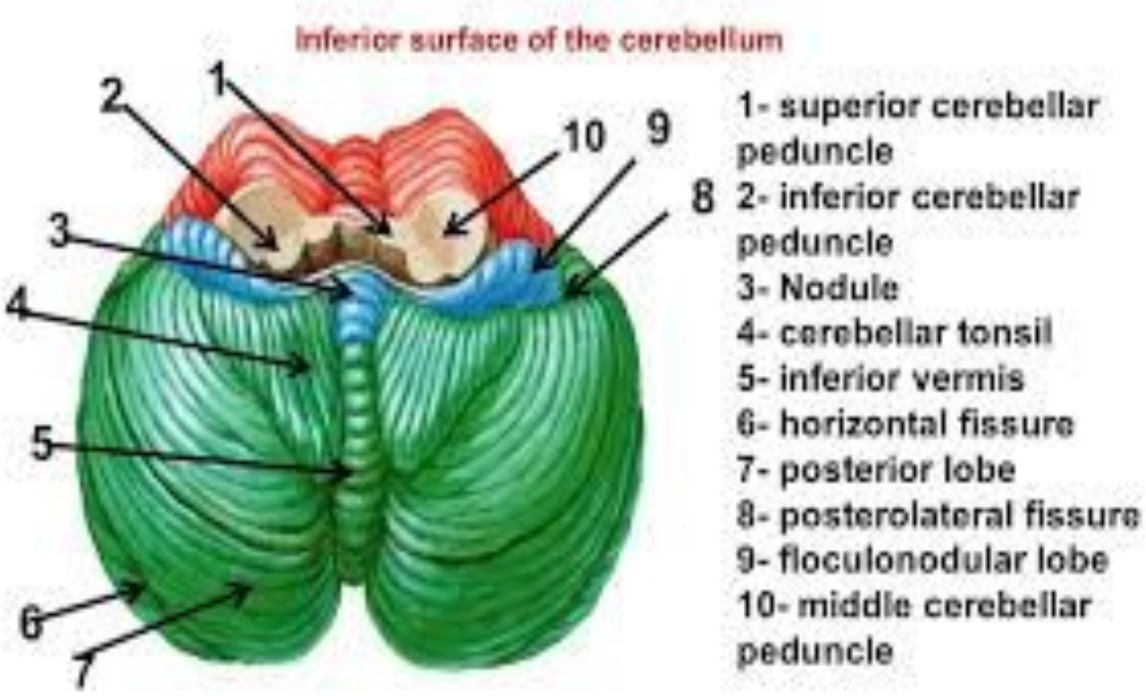
Lobes and fissures

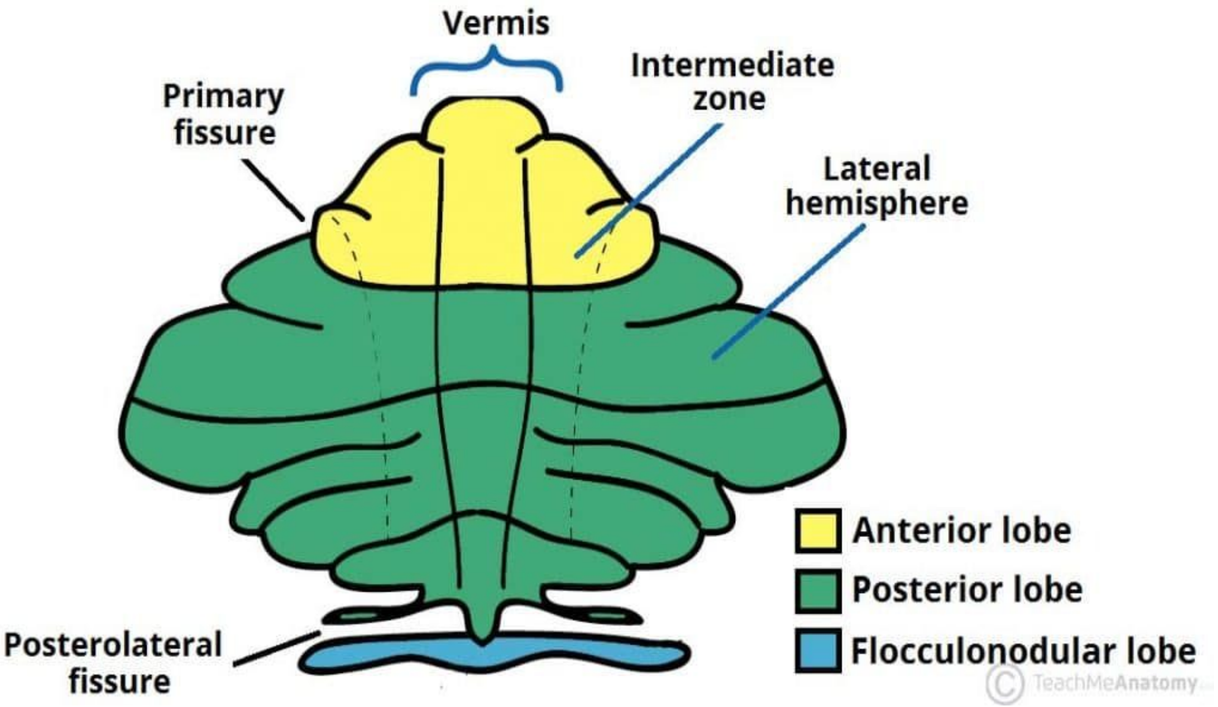
- It is divided into 3 main lobes: *anterior, middle,* and *flocculonodular lobes*.
- The anterior lobe is located superiorly.
- It is separated from the middle lobe by a V-shaped primary fissure.
- The middle lobe is the largest.
- It is situated between the primary and uvulonodular fissures.



- The flocculonodular lobe is situated posterior to the uvulonodular fissure.
- A deep horizontal fissure separates the superior surface from the inferior surface.

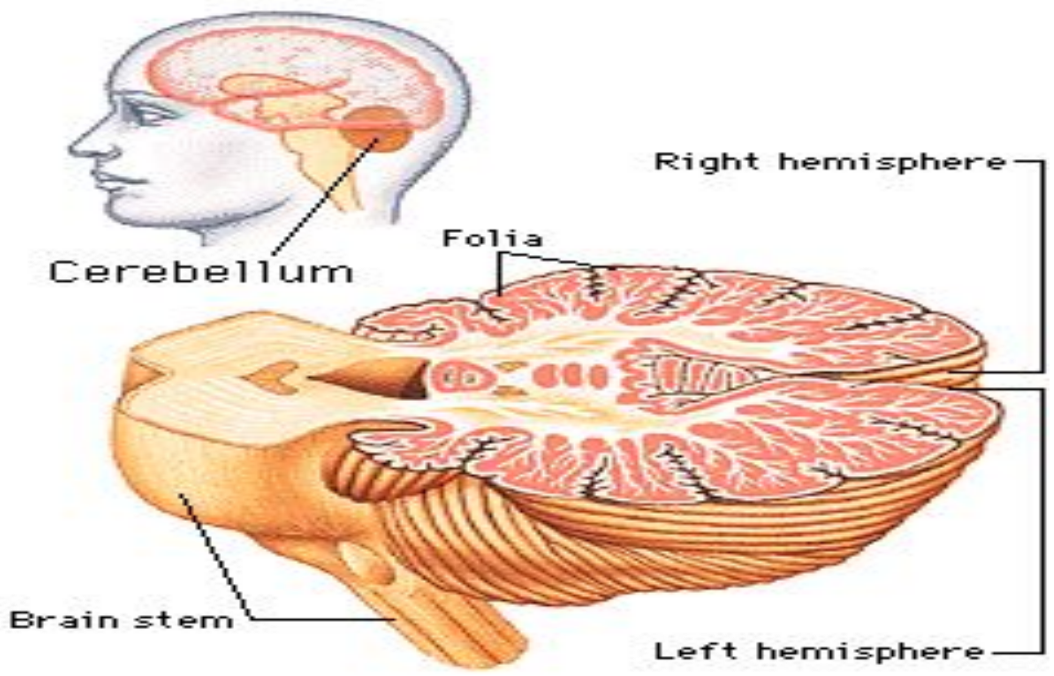


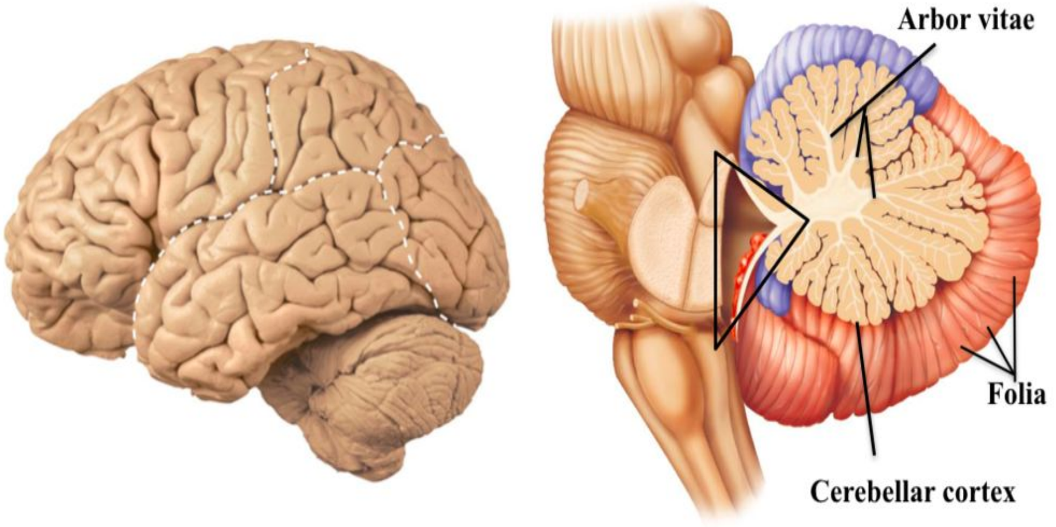




Internal structure

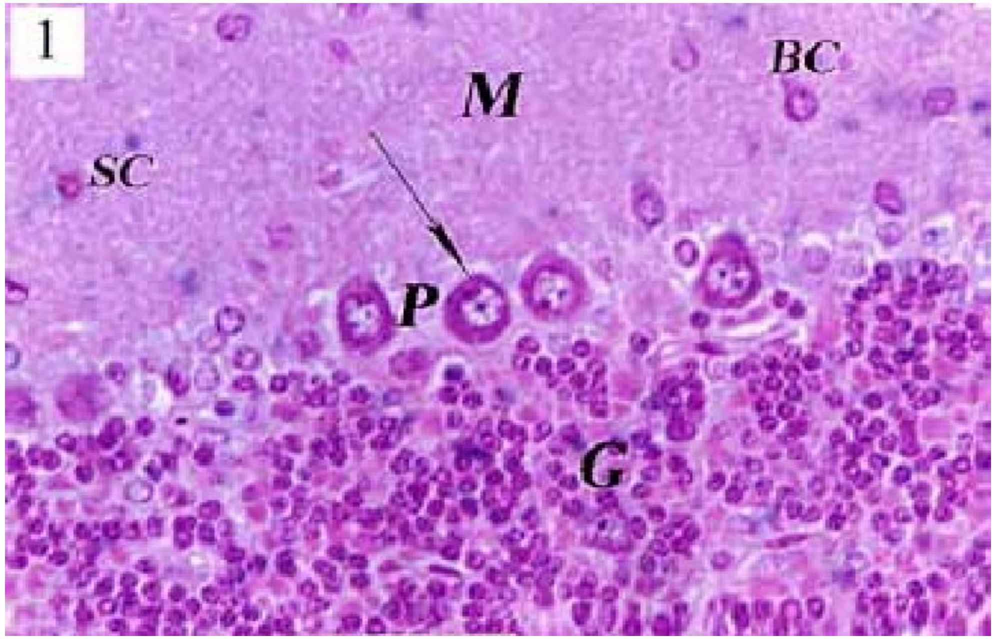
- Each cerebellar hemisphere contains an outer cortex of gray matter & inner white matter.
- Masses of gray matter are embedded in the white matter forming the *intracrebellar nuclei.*
- Each cerebellar cortex is thrown into numerous transverse folds known as *folia.*
- Internally each folium has a branched appearance called the *arbor vitae.*





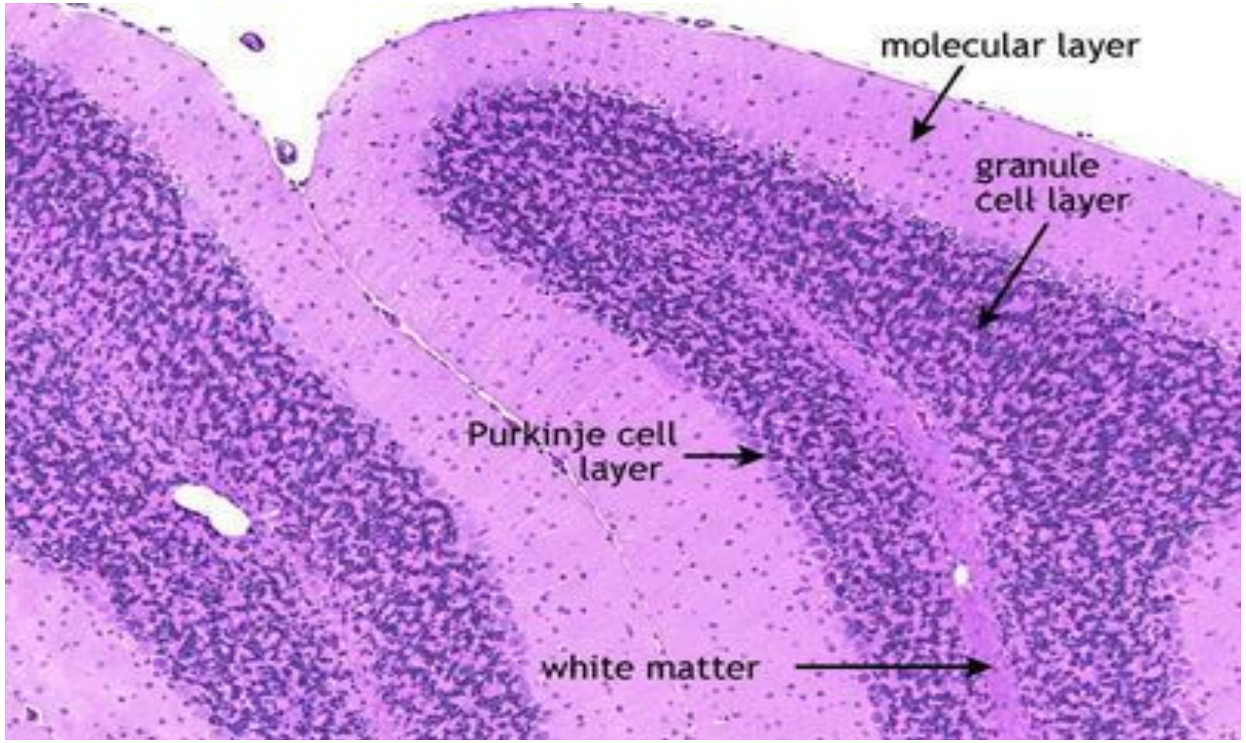
Histology of Cerebellar Cortex:

The section of cerebellar cortex shows three layers: *(1) Molecular layer, (2) Purkinje cell layer, & (3) Granular layer.*



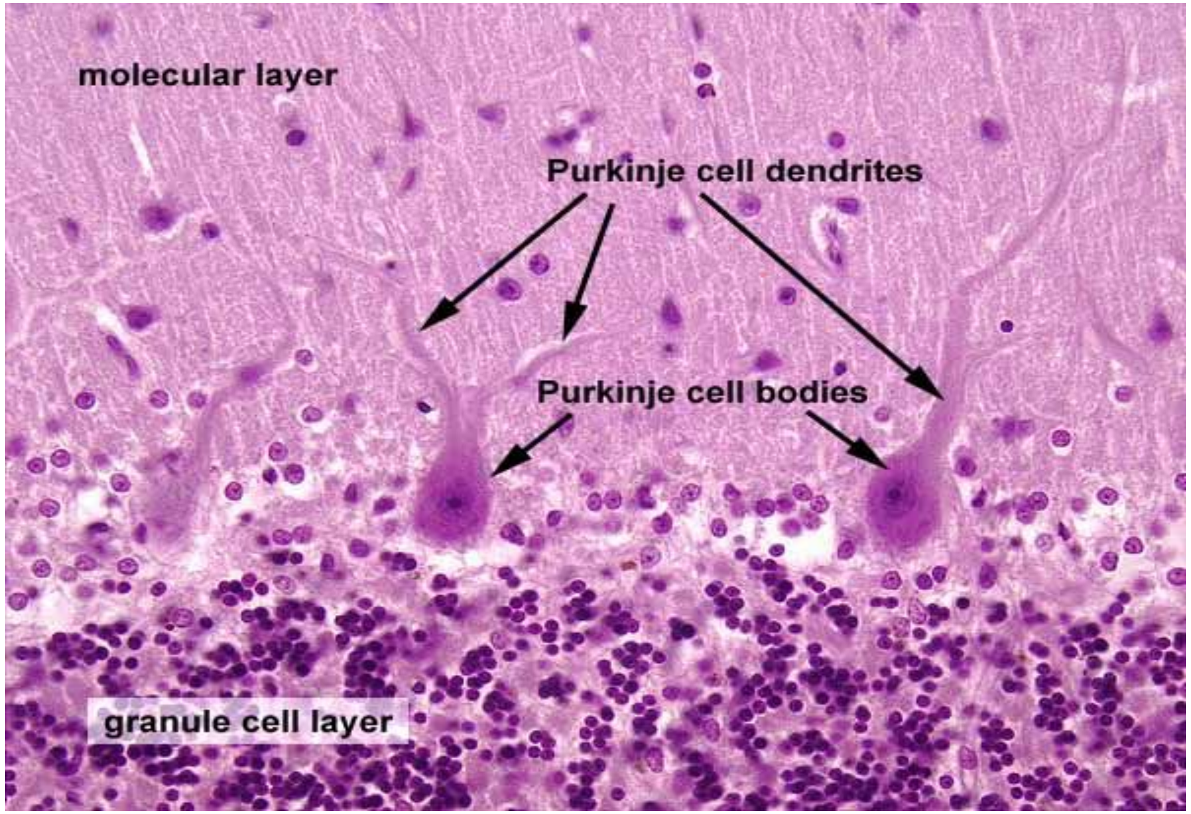
1. **Molecular layer:**

- It contains two types of cells, the outer stellate cells & the inner basket cells.
- This layer is a synaptic zone containing the dendritic arborization of Purkinje cells.
- It also contains many unmyelinated axons which run parallel to the surface of the cortex.



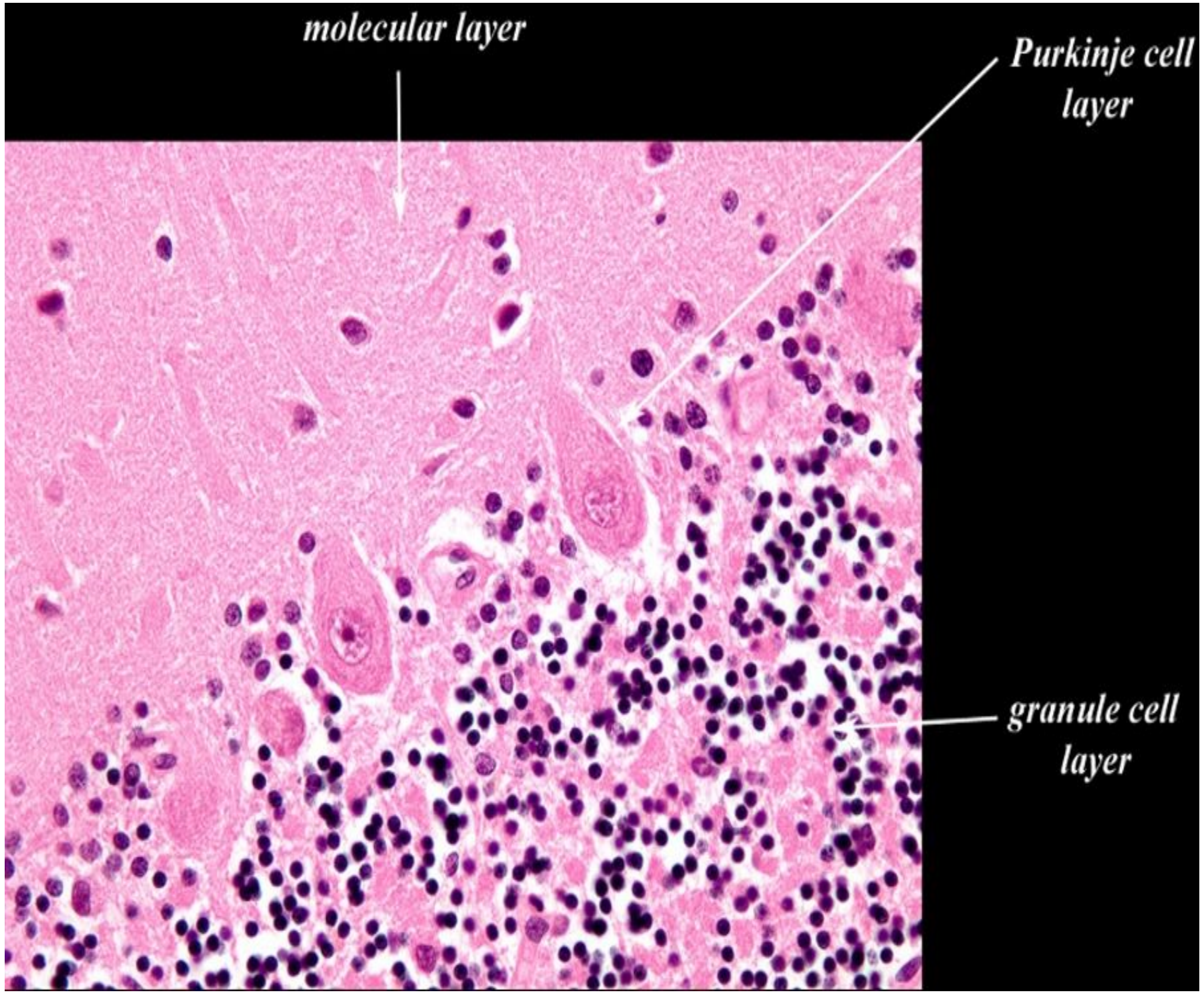
2. Purkinje cell layer:

- This layer comprises of a row of Purkinje cells.
- Purkinje cells are large multipolar neurons with flask-shaped cell bodies.
- Each Purkinje cell contains a vesicular nucleus with a prominent nucleolus and abundant Nissl granules in the cytoplasm.
- 2 to 3 primary dendrites arise from the apex of the cell body, enter the molecular layer and arborize.
- A single axon of each Purkinje cell arises from its base and passes through the granular layer to enter the white matter.
- Most axons of Purkinje end up in the deep cerebellar nuclei.

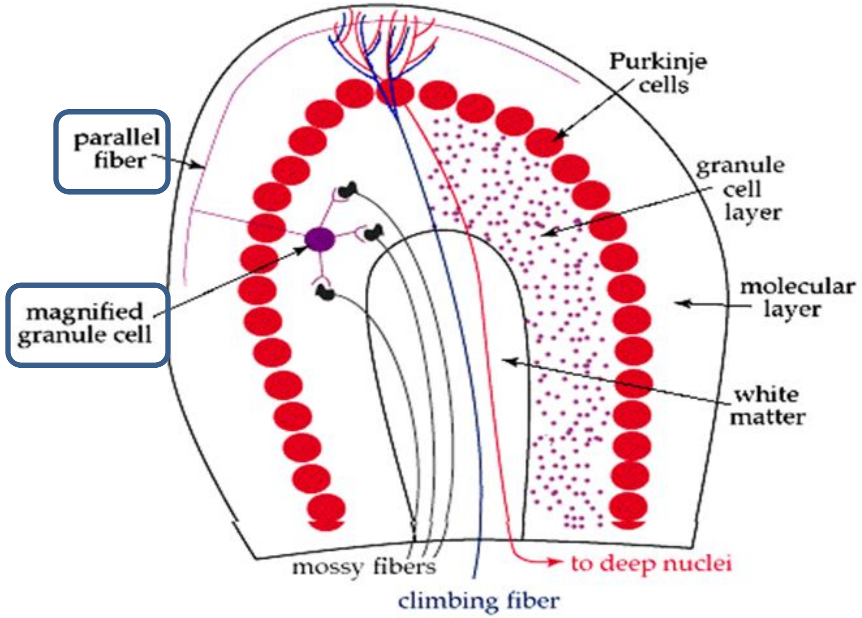


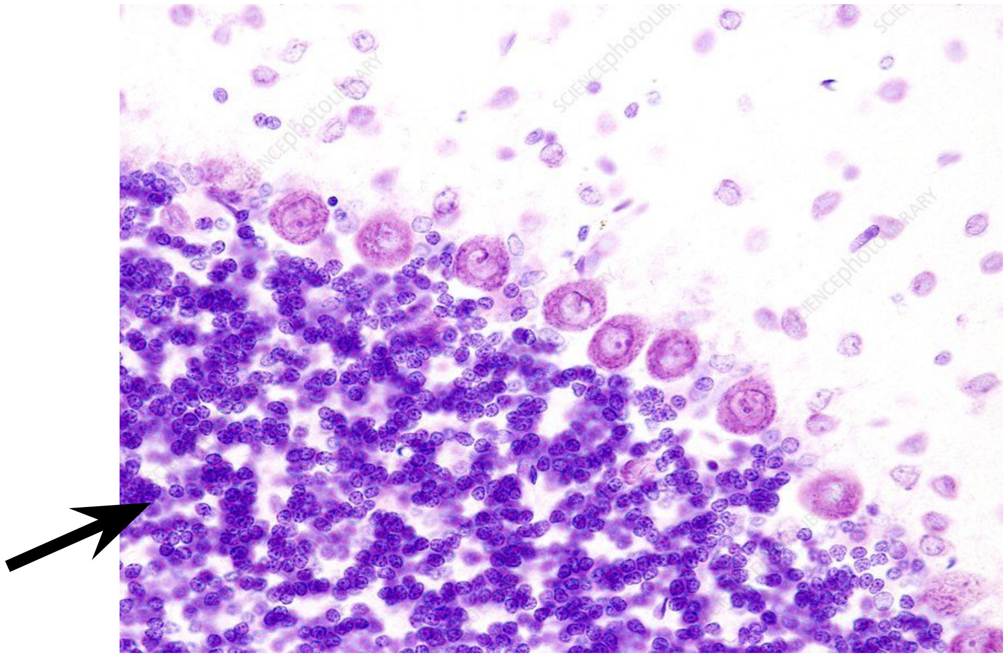
3. Granular layer:

- This layer is densely packed with small cells having deeply staining nuclei and scanty cytoplasm.
- The cytoplasm does not stain well & this layer appeared to contain closely packed deeply basophilic nuclei.
- Glomeruli/cerebellar islands are lighter staining areas randomly scattered in this layer.



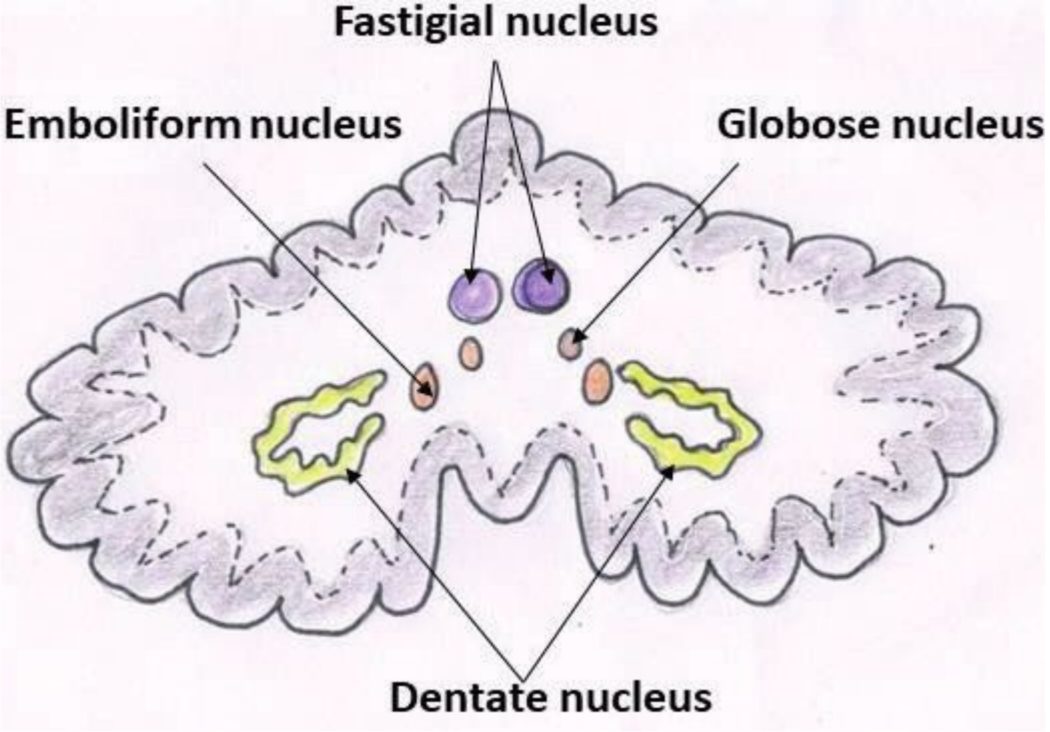
- Each granule cell usually has 4-5 small dendrites and a single unmyelinated axon.
- The axon ascends to the molecular layer and bifurcates in a T-shaped manner.
- These branches are called parallel fibers as they run parallel to the surface of cortex.
- These parallel fibers make synapse with the dendrites of Purkinje cells.

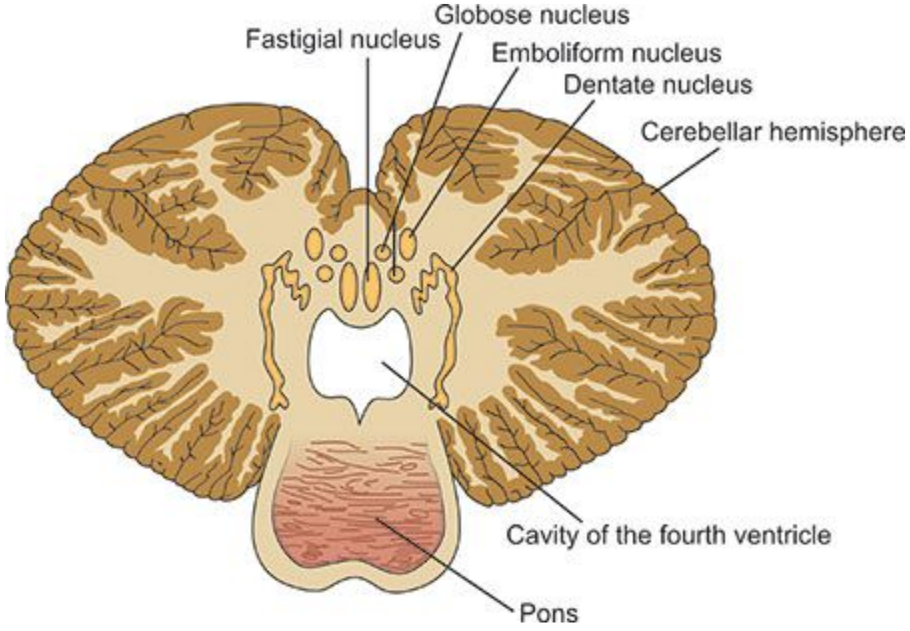




Intracerebellar (deep cerebellar) nuclei:

- These are the four masses of gray matter embedded in the white matter of cerebellum.
- In each cerebellar hemisphere, they are located from lateral to medial side as *dentate, emboliform, globose, & fastigial.*
- The dentate nucleus is the largest with the shape of a crumpled bag.
- These deep cerebellar nuclei consist of large multipolar neurons which form the cerebellar outflow.

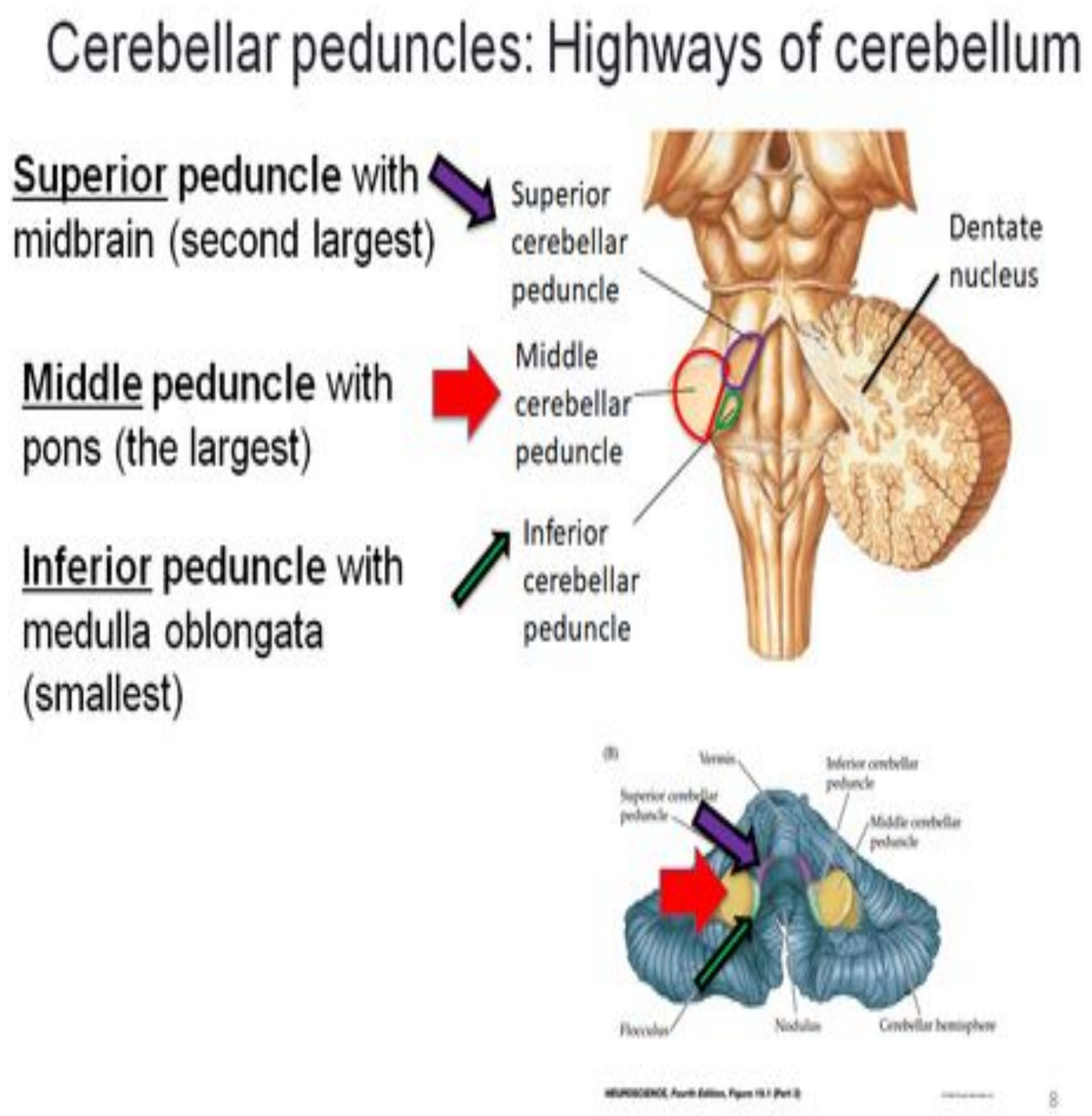




Cerebellar Peduncles:

There are *3* cerebellar peduncles:

1) **Superior cerebellar peduncle:** connects with midbrain and contains mostly efferent fibers.
2) **Middle cerebellar peduncle:** connects with pons and contains mostly afferent fibers.
3) **Inferior cerebellar peduncle:** connects with medulla & spinal cord and contains both afferent & efferent fibers.

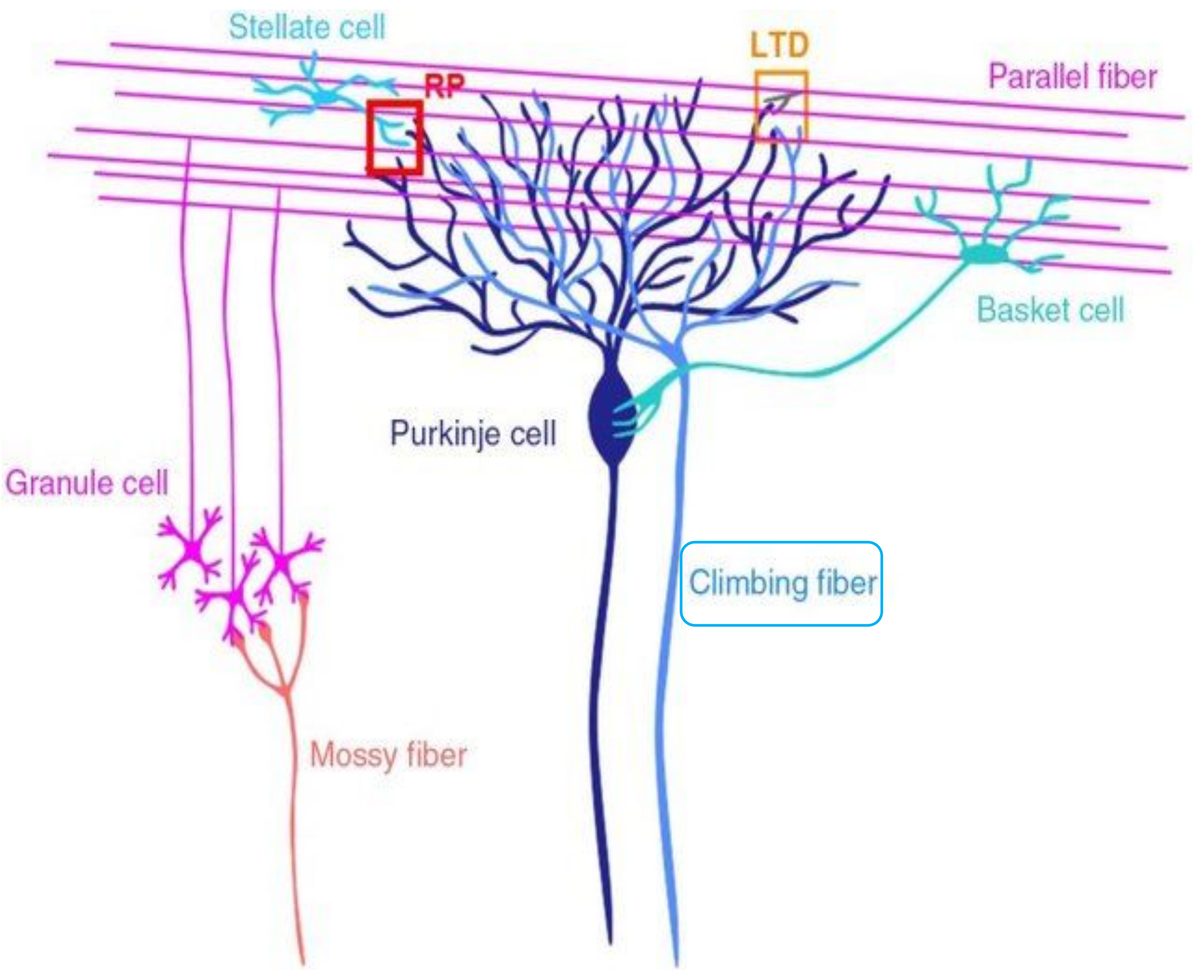


Cerebellar Afferent Fibers:

Afferents to the cerebellum have two types of endings:

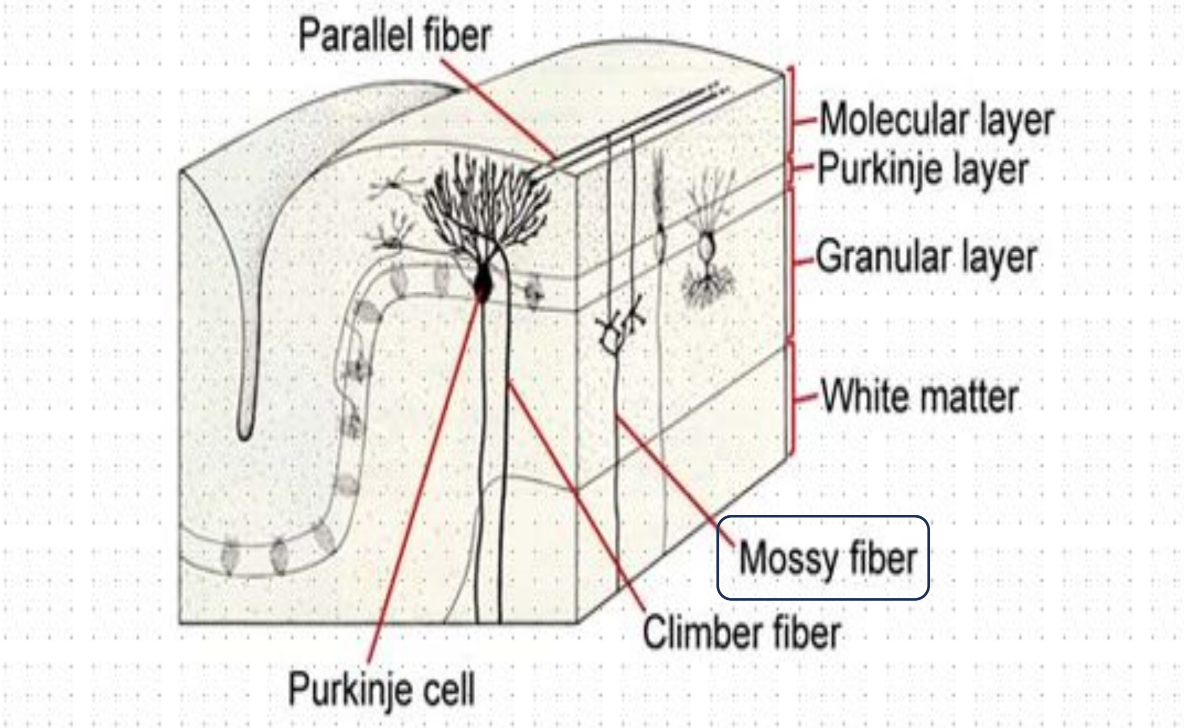
i. Climbing fibers:

- They are actually the terminal fibers of the olivo-cerebellar neurons
- They climb through granular layer.
- They end in a tuft of terminal branches in the molecular layer.
- They synapse with the dendrites of Purkinje cells in the ratio of 1:1.
- They secrete glutamate as neurotransmitter.



ii. Mossy fibers:

- They are the terminal fibers of all other cerebellar afferent tracts.
- They branch and terminate in the granular layer.
- They have dilated endings (rosettes).
- They synapse with the dendrites of granule cells and axons of Golgi type II neurons.
- They serve as inhibitory interneurons.



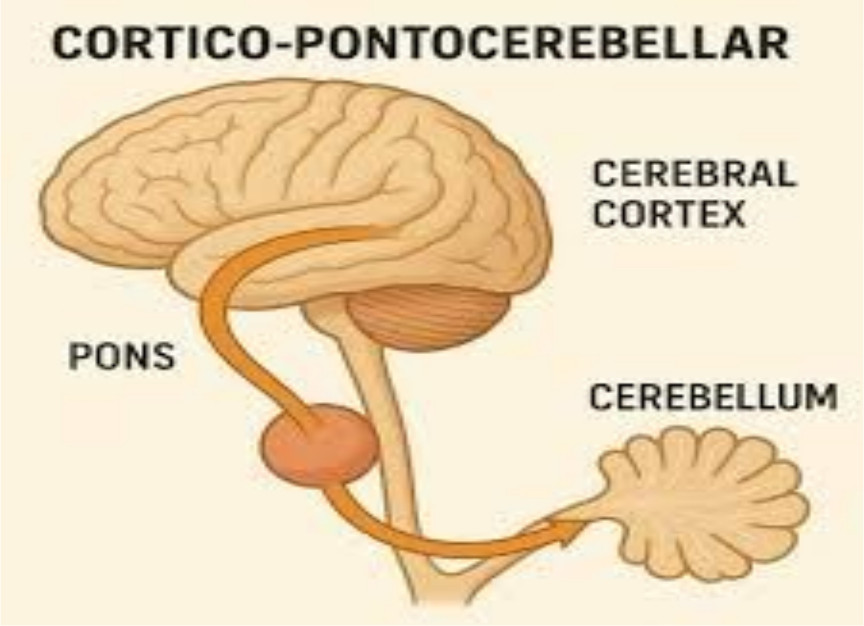
Cerebellar Afferent Pathways:

The cerebellum receives afferent information from the cerebral cortex by 3 pathways:

I. *Cortico-ponto-cerebellar pathway*
II. *Cerebro-olivo-cerebellar pathway*
III. *Cerebro-reticulo-cerebellar pathway*

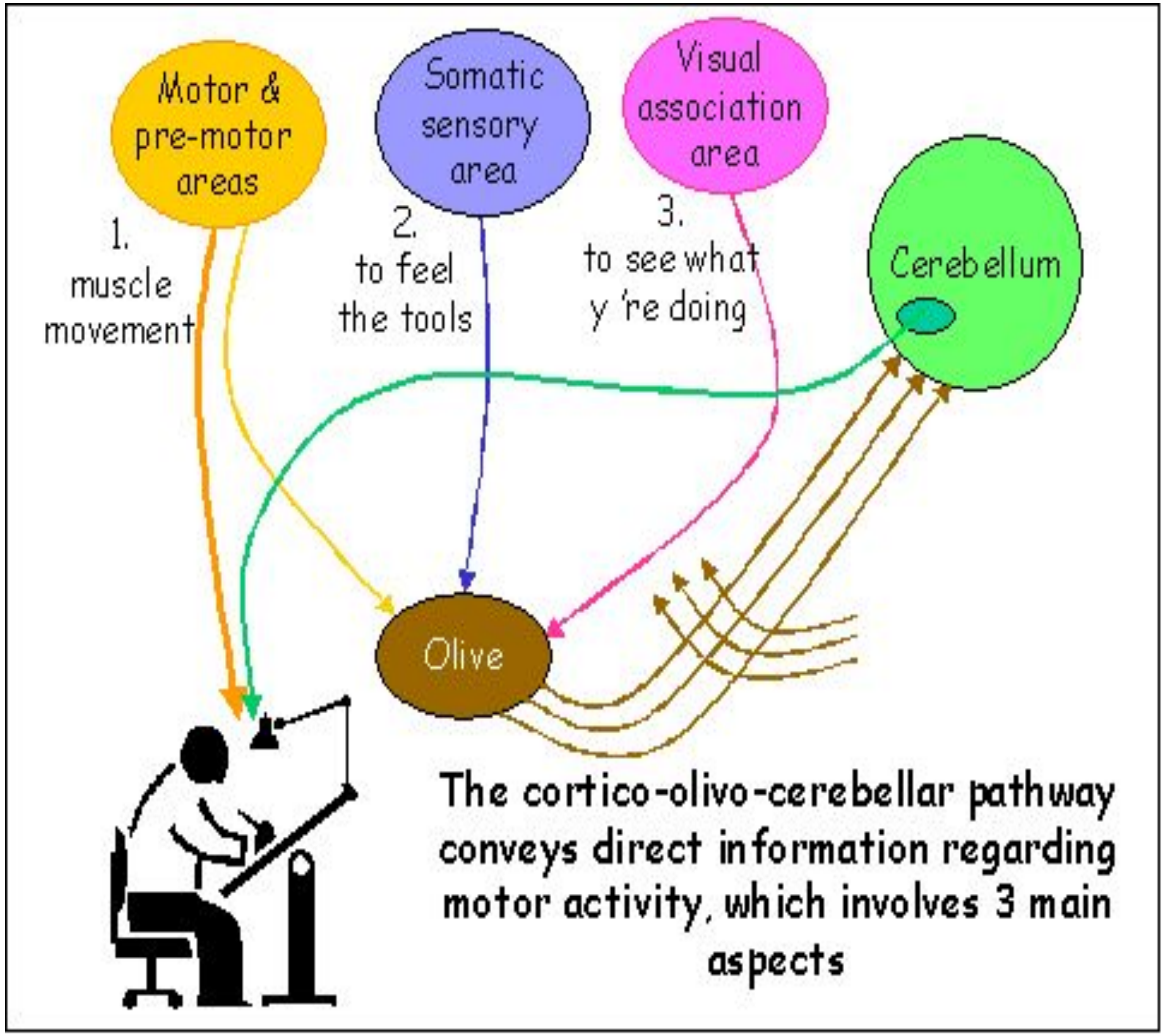
I. Cortico-ponto-cerebellar pathway:

- Firstly, the corticopontine fibers arise in the frontal, parietal, temporal, & occipital lobes.
- These fibers descend through the corona radiata & internal capsule to end up in the pontine nuclei.
- Secondly, the pontine nuclei give rise to the transverse pontine fibers.
- The transverse pontine fibers cross midline and enter the opposite cerebellar hemisphere via middle cerebellar peduncle.



II. Cerebro-olivo-cerebellar Pathway:

- Firstly, the corticopontine fibers arise in the frontal, parietal, temporal, & occipital lobes.
- These fibers descend through the corona radiata & internal capsule to end up bilaterally in the inferior olivary nuclei.
- Secondly, the fibers which arise from the inferior olivary nuclei cross midline to end up in the opposite cerebellar hemisphere via inferior cerebellar peduncle.
- These terminal fibers are called climbing fibers of the cerebellar cortex.



Cerebellar Afferent Fibers from the Spinal Cord:

Spinocerebellar tracts ventral &dorsal

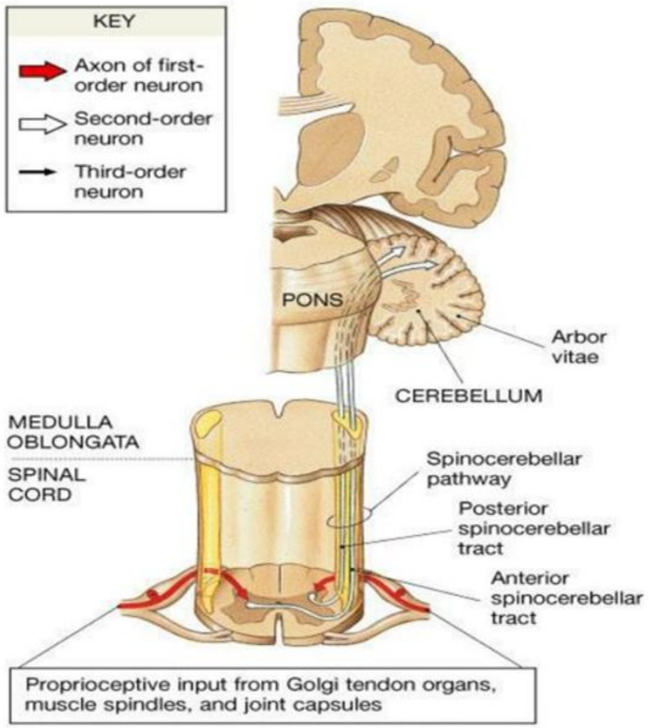
❖Carry information from muscle spindle, Golgi tendon organ and tactile receptors to the cerebellum.

❖To control posture and coordination of movement.

❖The dorsal root ganglion is 1^{st} order neuron.

❖The 2^{nd} order neuron cell bodies lie in the base of dorsal horn (Clark's) .

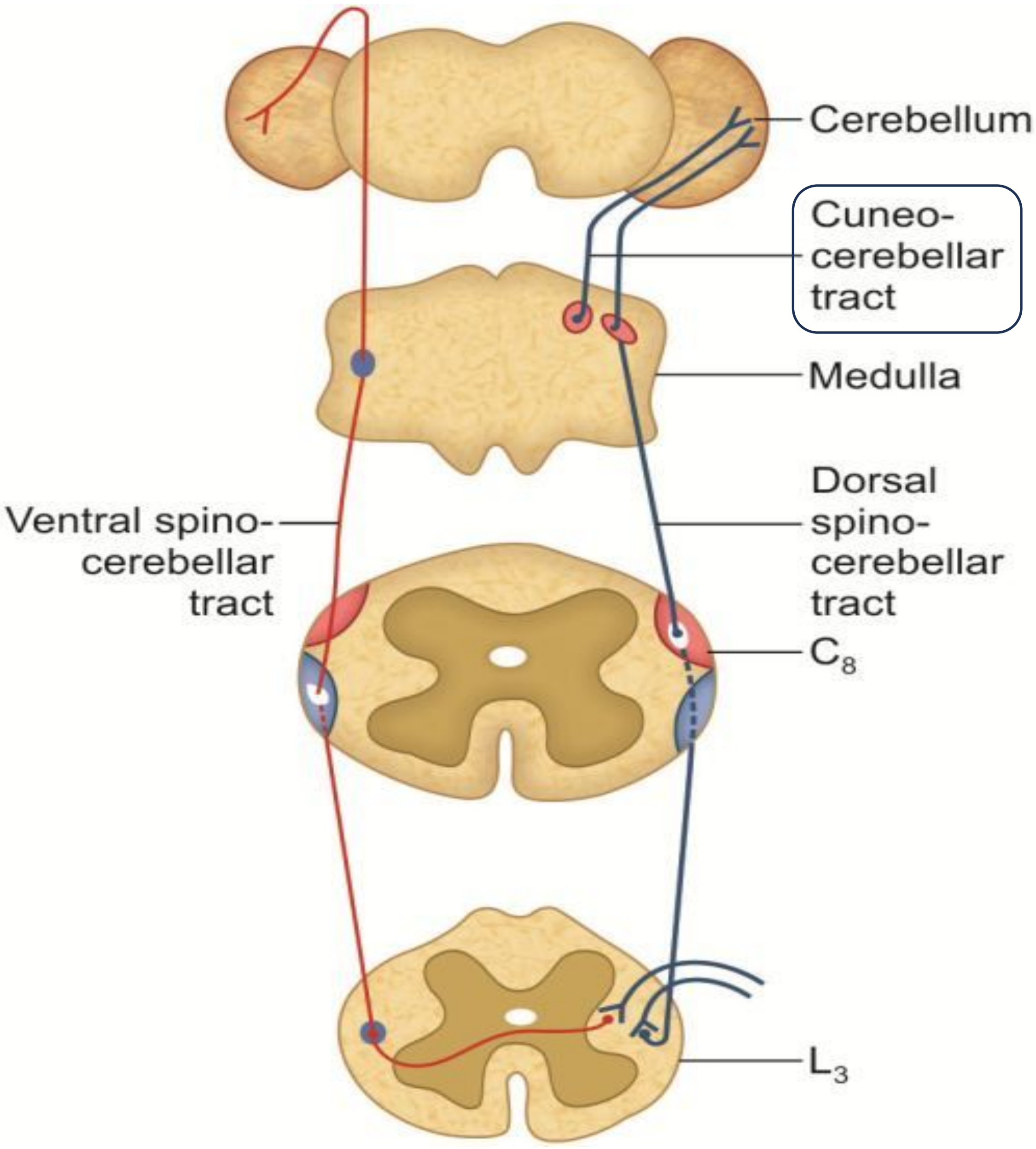
❖Terminate directly in the cerebellar cortex (vermis)



Muscle joint sense pathways to cerebellum				
Pathways	**Sensation**	**First-order neuron**	**Second-order neuron**	**Destination**
Anterior and posterior spinocerebellar	Unconscious muscle joint sense	Posterior root ganglion	Nucleus dorsalis	Cerebellar cortex

Cuneo-cerebellar Tract:

- These fibers arise from the nucleus cuneatus of medulla oblongata.
- They are known as posterior external arcuate fibers.
- They enter the ipsilateral cerebellar hemisphere through inferior cerebellar peduncle.
- They terminate in the cerebellar cortex as mossy fibers.
- They convey information of muscle-joint sense to the cerebellum.



Cerebellar Efferent Pathways:

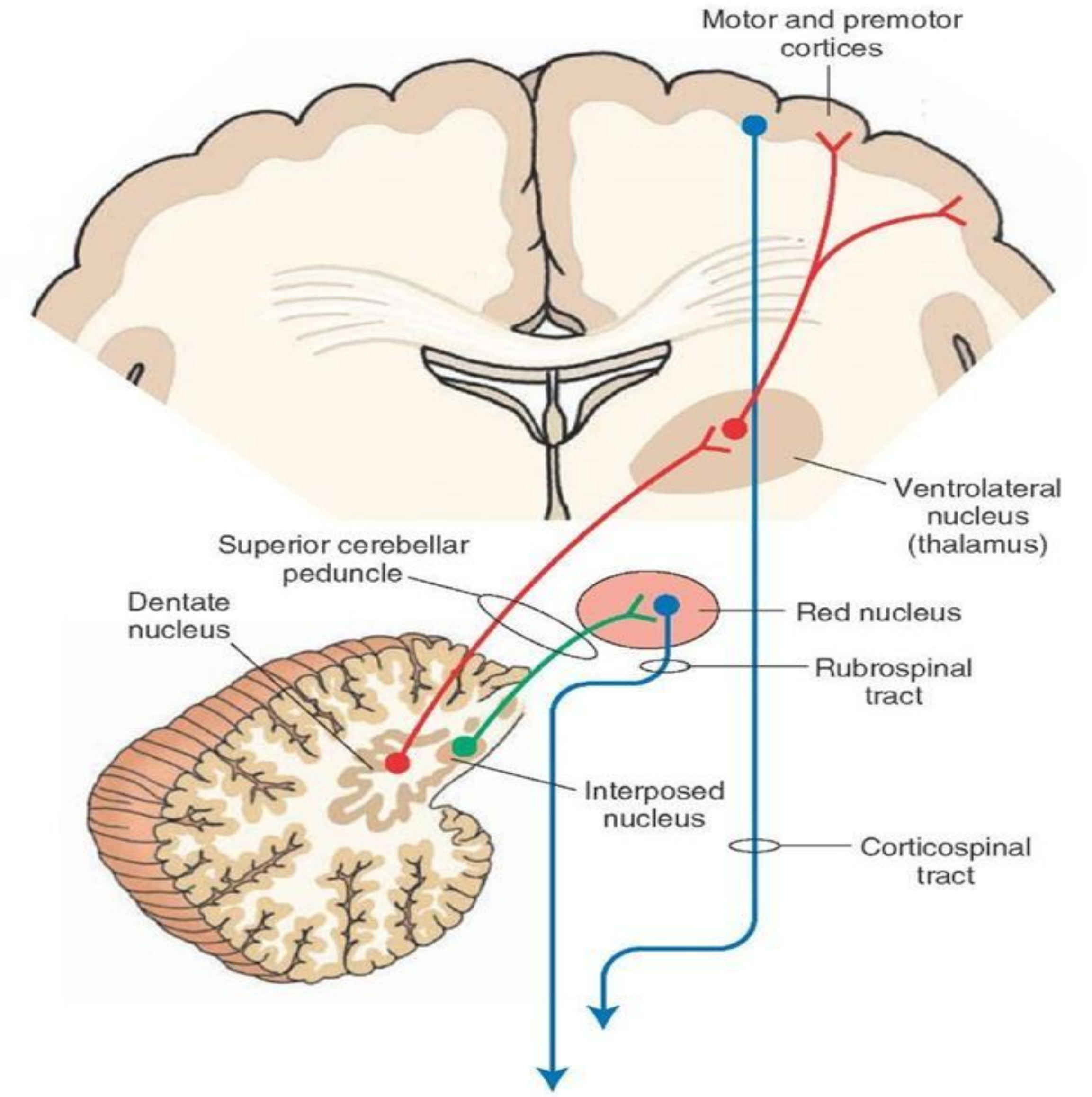
- The axons of the neurons that end up in the deep cerebellar nuclei, form the efferent outflow of the cerebellum.
- The entire outflow of the cerebellar cortex is through the axons of Purkinje cells.
- The efferent fibers link cerebellum with the red nucleus, thalamus, vestibular complex, & reticular formation.

The efferent pathways of cerebellum are as follows:

I. *Dento-Emboliform-Rubral Pathway*
II. *Dentothalamic Pathway*
III. *Fastigial Vestibular Pathway*
IV. *Fastigial Reticular Pathway*

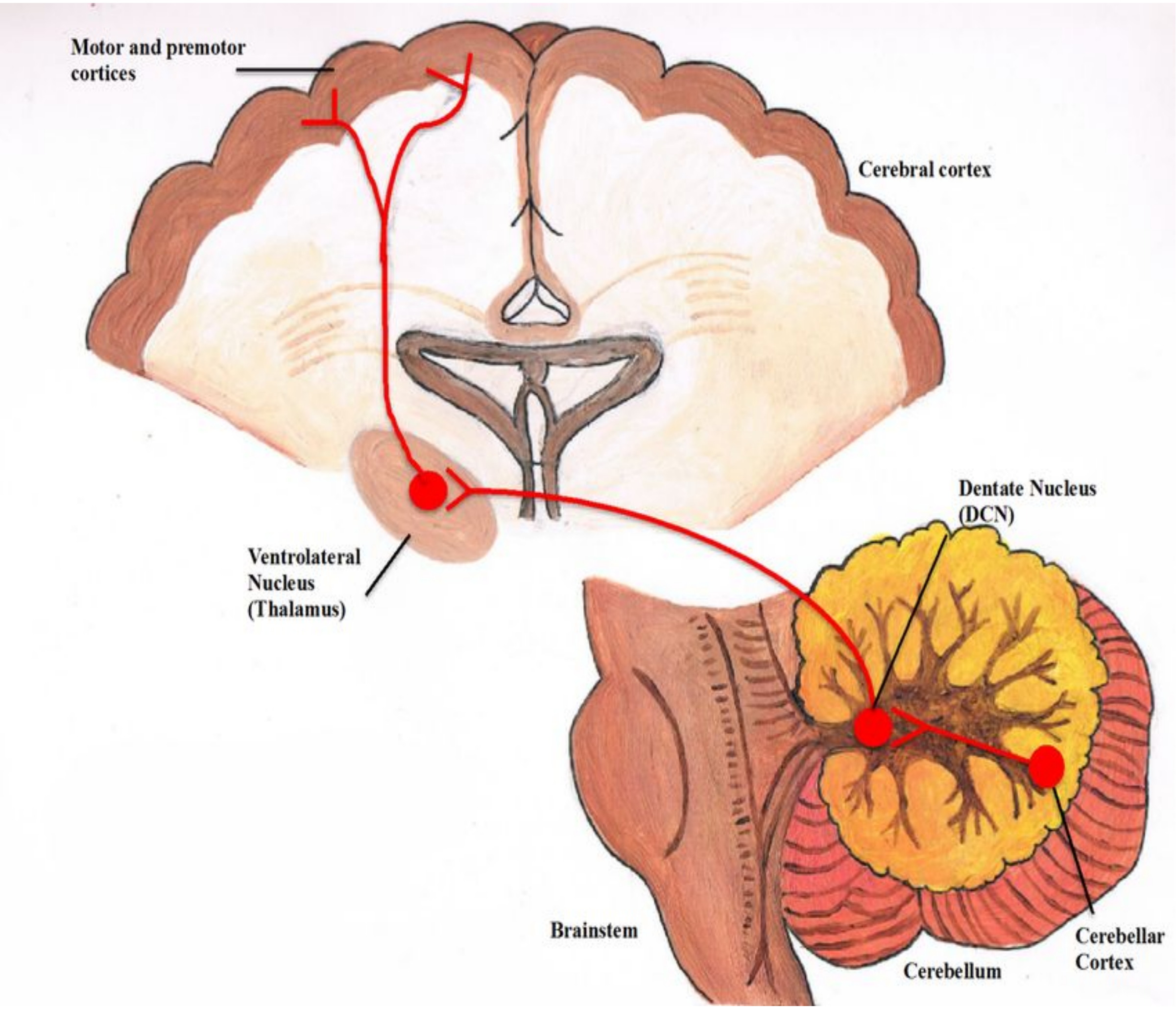
I. Globose-Emboliform-Rubral Pathway:

- Its fibers arise from the globose and emboliform nuclei and pass through the superior cerebellar peduncle.
- The fibers cross the midline in the decussation of superior cerebellar peduncles to reach the red nucleus of the opposite side.
- The red nucleus gives rise to the rubrospinal tract.
- In this way the globose and emboliform nuclei effects motor activity on the same side of the body.



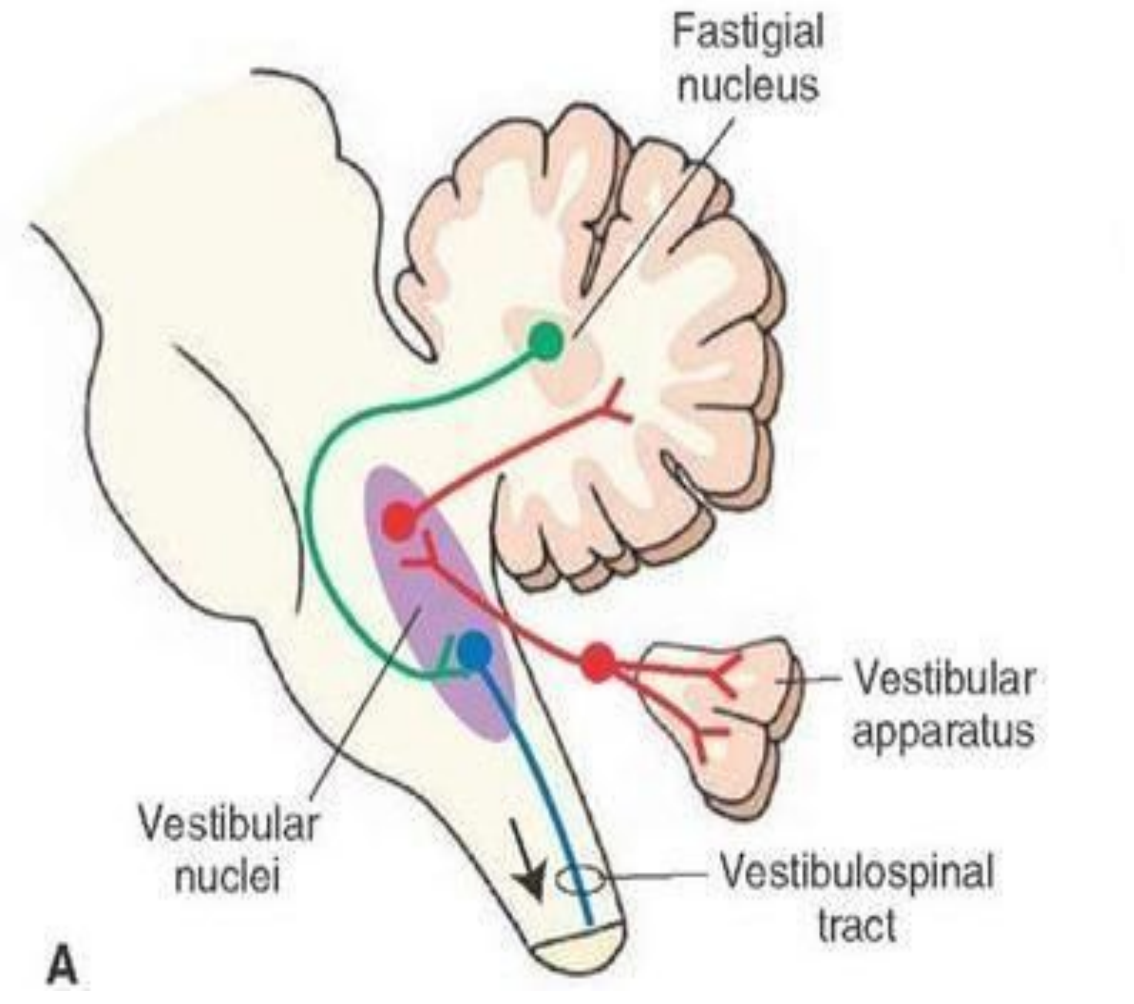
II. Dento-thalamic Pathway:

- Its fibers arise from the dentate nucleus and pass through the superior cerebellar peduncle.
- The fibers cross the midline in the decussation of superior cerebellar peduncles to reach the venterolateral (VL) nucleus of the thalamus.
- The axons of these thalamic neurons ascend via the internal capsule & corona radiata to reach the primary motor area of the cerebral cortex.
- In this way the dentate nucleus may influence motor activity by acting on the motor neurons of the opposite cerebral cortex.



III. Fastigial Vestibular Pathway:

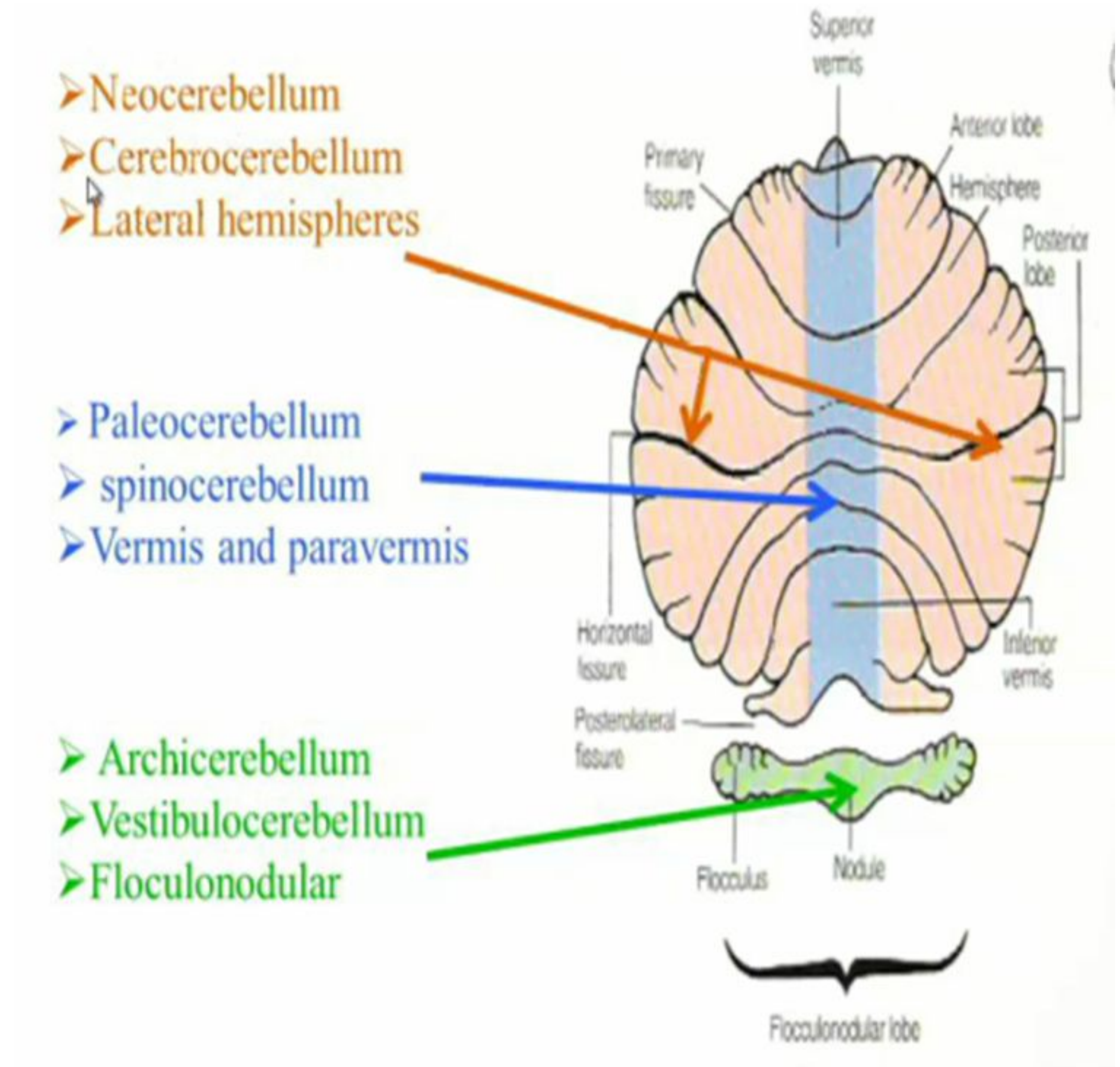
- It begins in the fastigial nucleus, and passes through the inferior cerebellar peduncle.
- Then ends up in the lateral vestibular nucleus of both sides.
- The lateral vestibular nucleus gives rise to the vestibulospinal tract.
- Hence the fastigial nucleus exerts a facilitatory influence mainly on the ipsilateral extensor muscle tone.



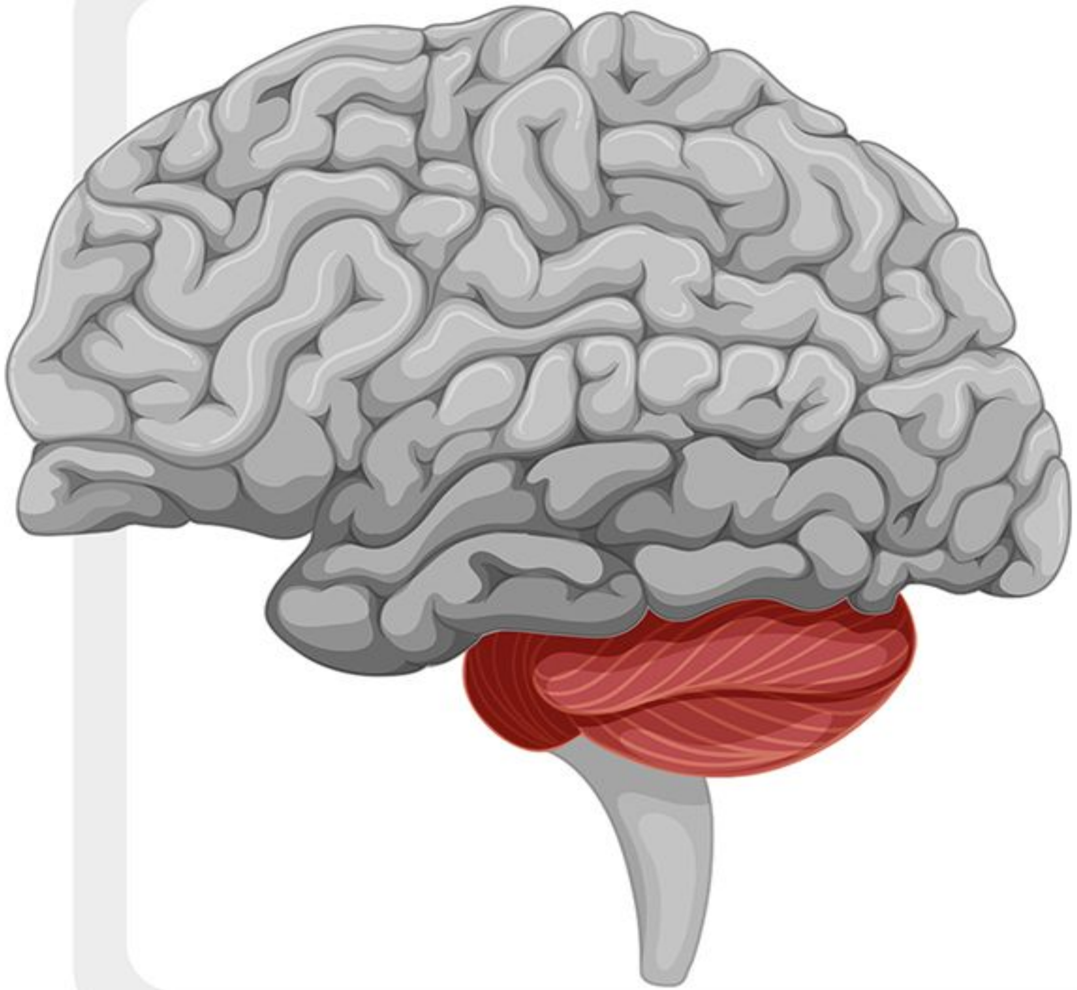
Functional Divisions of Cerebellum:

The *paleocerebellum, archicerebellum, and neocerebellum* are the 3 phylogenetically distinct regions of the cerebellum, each with a specific function and evolutionary origin:

1. The **archicerebellum** (vestibulocerebellum) is the oldest part, responsible for balance and eye movements.
2. The **paleocerebellum** (spinocerebellum) is the intermediate region, which regulates muscle tone, posture, and the coordination of crude limb movements.
3. The **neocerebellum** (cerebrocerebellum) is the newest and largest part, involved in planning and the smooth execution of skilled, voluntary movements.



CEREBELLUM FUNCTION



The cerebellum is responsible for coordinating voluntary movement, balance, and posture.

It receives information from sensory systems and the spinal cord, and sends out instructions to the muscles to control movement.

The cerebellum also plays a role in cognitive functions such as attention, language, and working memory. Overall, it is a crucial component of the brain that helps us move smoothly and maintain stability in our daily activities.

CEREBELLUM = "LITTLE BRAIN"

CEREBELLUM

* COORDINATES MOVEMENTS
* CONTROLS POSTURE, BALANCE & FINE MOTOR MOVEMENT
* INVOLVED IN MOTOR LEARNING

Applied Anatomy:

- A lesion in one cerebellar hemisphere gives rise to signs & symptoms on the same side of the body.
- Few signs & symptoms of cerebellar dysfunctions are as follows:

i. **Hypotonia:** in this, the muscles lose resistance to palpation or muscle tone is decreased.

ii. **Postural changes:** head is rotated & flexed and shoulder on the side of the lesion is lower than on the normal side.

iii. **Ataxia:** disturbances of voluntary movements.

iv. **Dysdiadochokinesia:** it is the inability to perform alternating movements regularly & rapidly.

References:

CLINICAL NEUROANATOMY

Richard S. Snell

8th edition

Chapter 6

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